From Interpolation to Extrapolation:
A Minimal Resonance-Based Experiment Inspired by I2OS

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Abstract

This paper investigates extrapolation capability in low-data regimes through a minimal resonance-based experiment.

Conventional machine learning models are effective at interpolation but often fail in extrapolation, especially when data is limited and the target function contains complex structure.

We compare a simple linear baseline with a resonance-based feature model that encodes periodic structure. Results show that the resonance model preserves structural behavior and achieves significantly lower error in the extrapolation region.

These findings support the I2OS perspective that intelligence should be understood as structure-preserving resonance rather than data fitting.

1 Introduction

Most machine learning systems rely on large datasets, optimization, and predictive accuracy. However, this paradigm breaks down under low-data conditions, unknown domains, and complex nonlinear structures.

Recent studies in quantum circuit learning suggest that complex structures can be learned even with limited data, enabling extrapolation beyond the training domain.

This paper proposes a minimal interpretation: extrapolation becomes possible when structural resonance is encoded.

2 I2OS Perspective

In I2OS, intelligence is defined as:

$$I = (\nabla M) \otimes R \tag{1}$$

where ∇M represents the meaning gradient and R represents resonance.

3 Method

3.1 Target Function

$$y = \sin(x) + 0.3\sin(3x) \tag{2}$$

3.2 Data Setup

Training domain:

$$x \in [0, 6]$$

Extrapolation domain:

$$x \in [6, 10]$$

3.3 Models

3.3.1 Baseline Model

$$\hat{y} = ax + b \tag{3}$$

3.3.2 Resonance Model

$$\phi(x) = [\sin x, \cos x, \sin 2x, \cos 2x, \sin 3x, \cos 3x] \tag{4}$$

3.4 Evaluation

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2 \tag{5}$$

4 Results

Figure 1 shows that the baseline model diverges immediately outside the training region, while the resonance model preserves the structure of the target function.

Figure 2 shows that the baseline model accumulates large error, whereas the resonance model maintains low and stable error in the extrapolation region.

5 Discussion

This experiment shows that extrapolation depends not on data quantity but on structural compatibility between model and system.

The resonance model succeeds because its feature space aligns with the underlying periodic structure.

This observation is consistent with findings in quantum circuit learning, where strong generalization is achieved under limited data conditions.

This result suggests that extrapolation capability is fundamentally governed by structural alignment rather than dataset scale.

6 Conclusion

We demonstrate that resonance-based representations enable extrapolation under low-data conditions.

> Extrapolation depends on structure, not data volume.

7 Appendix

Code for the experiment is provided for reproducibility.

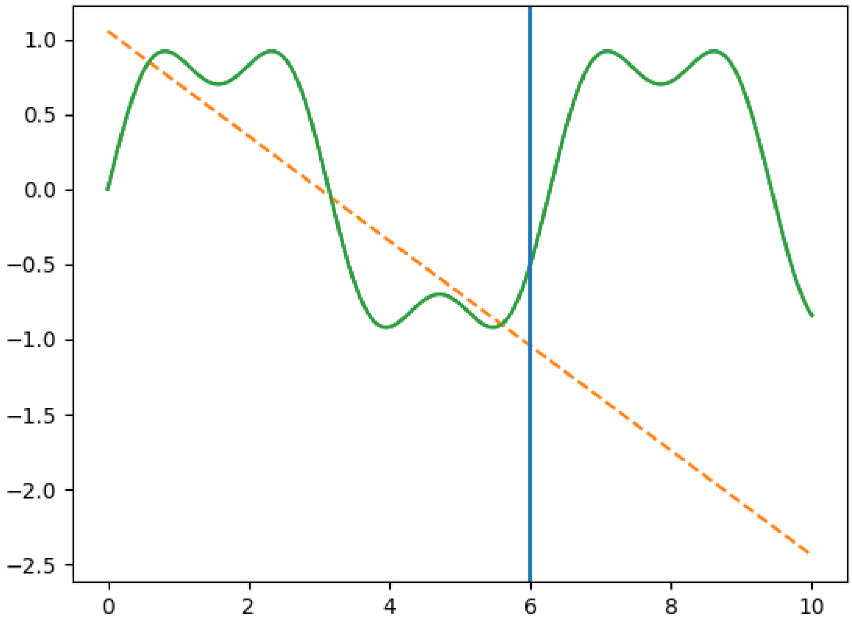

Figure 1: Extrapolation behavior of baseline and resonance models

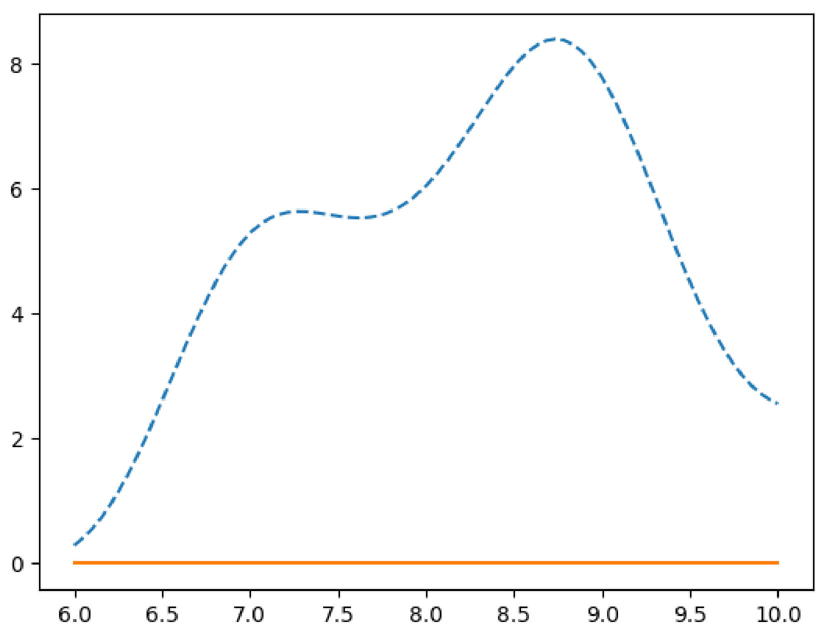

Figure 2: Squared error in the extrapolation region